# PRACTICE SHEET

1. An equilateral triangle is inscribed in the circle $x^2 + y^2 = a^2$ with one of the vertices at (a,0). What is the equation of the side opposite to this vertex?
(a) $2x - a = 0$ (b) $x + a = 0$
(c) $2x + a = 0$ (d) $3x - 2a = 0$

2. What is the radius of the circle passing through the points (0, 0) (a, 0) and (0, b)?
(a) $\sqrt{a^2 - b^2}$ (b) $\sqrt{a^2 + b^2}$
(c) $\frac{1}{2}\sqrt{a^2 + b^2}$ (d) $2\sqrt{a^2 + b^2}$

3. If two circles A, B of equal radii pass through the centres of each other, then what is the ratio of the length of the smaller arc to the circumference of the circle A cut off by the circle B?
(a) 1/2 (b) 1/4
(c) 1/3 (d) 2/3

4. What is the equation of a circle, whose centre lies on the x – axis at a distance h from the origin and the circle passes through the origin?
(a) $x^2 + y^2 - 2hx = 0$
(b) $x^2 + y^2 - 2hx + h^2 = 0$
(c) $x^2 + y^2 + 2hxy = 0$
(d) $x^2 + y^2 - h^2 = 0$
(c) $x^2 + y^2 - 2x - 2y + 1 = 0$
(d) None of these

7. If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ $(c > 0)$ touches the y – axis, then which one of the following is correct?
(a) $g = -\sqrt{c}$ only (b) $g = \pm\sqrt{c}$
(c) $f = \sqrt{c}$ only (d) $f = \pm\sqrt{c}$

8. The circle $x^2 + y^2 + 4x - 4y + 4 = 0$ touches.
(a) Only the X-axis (b) Only the Y-axis
(c) Both the axis (d) Neither of the axis

9. The equation of the circle which touches the axes at a distance 5 from the origin is $y^2 + x^2 - 2\alpha x - 2\alpha y + \alpha^2 = 0$. What is the value of $\alpha$?
(a) 4 (b) 5
(c) 6 (d) 7

10. The equation of the circle passing through the origin is $x^2 + y^2 - 6x + 2y = 0$. The equation of one of its diameters is:
(a) $x + 3y = 0$ (b) $x + y = 0$
(c) Both (a) and (b) (d) $3x + y = 0$

11. What is the equation to circle which touches both the axes and has centre on the line $x+y = 4$?